University of Asia Pacific

Department of Computer Science and Engineering **CSE 437: Robotics Lab**

LAB REPORT

Experiment Number: 05

Experiment Title: Designing a Bluetooth-Based Android-Controlled Robot.

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1. Experiment Name

Designing a Bluetooth-Based Android-Controlled Robot.

2. Objective

The main objective of this experiment is to design and develop a Bluetooth-based Android controlled robot using Arduino UNO. The system enables wireless communication between an Android smartphone and a robot through a Bluetooth module, allowing the user to control robot movement remotely. This project helps in understanding motor driver interfacing, wireless communication, and real-time robotic control using embedded systems.

3. Apparatus / Hardware & Software Requirements

Hardware Components

- Arduino UNO
- HC-05 Bluetooth Module
- L293D Motor Driver IC
- 2 DC Gear Motors
- Breadboard
- Battery Pack (4 × AA Batteries)
- Jumper Wires
- Robot Chassis with Wheels
- Android Smartphone

Software Requirements

- Arduino IDE
- Android Bluetooth Controller App (Any Bluetooth RC Controller)
- USB Cable
- Tinkercad for Simulation

4. Circuit Diagram / Schematic

Main system connection:

- Arduino UNO controls motors through L293D.
- HC-05 Bluetooth module sends commands from Android phone.
- External battery powers motors.
- Arduino provides logic control signals.

Pin Configuration

HC-05 Bluetooth Module → Arduino

HC-05 Pin	Arduino Pin
VCC	5V
GND	GND
TX	RX (Pin 0)
RX	TX (Pin 1)

L293D Motor Driver → Arduino

L293D Pin	Connection
Pin 1 (Enable 1)	Arduino Pin 5
Pin 2 (IN1)	Arduino Pin 8
Pin 7 (IN2)	Arduino Pin 9
Pin 9 (Enable 2)	Arduino Pin 6
Pin 10 (IN3)	Arduino Pin 10
Pin 15 (IN4)	Arduino Pin 11
Pin 4,5,12,13	GND
Pin 16	5V (Logic Supply)
Pin 8	Battery + (Motor Supply)

Motor Connections

- Motor A → L293D Output 1 & 2
- Motor B → L293D Output 3 & 4

Power Connections

- Arduino powered via USB.
- Motors powered by AA battery pack.
- All GND connected together.

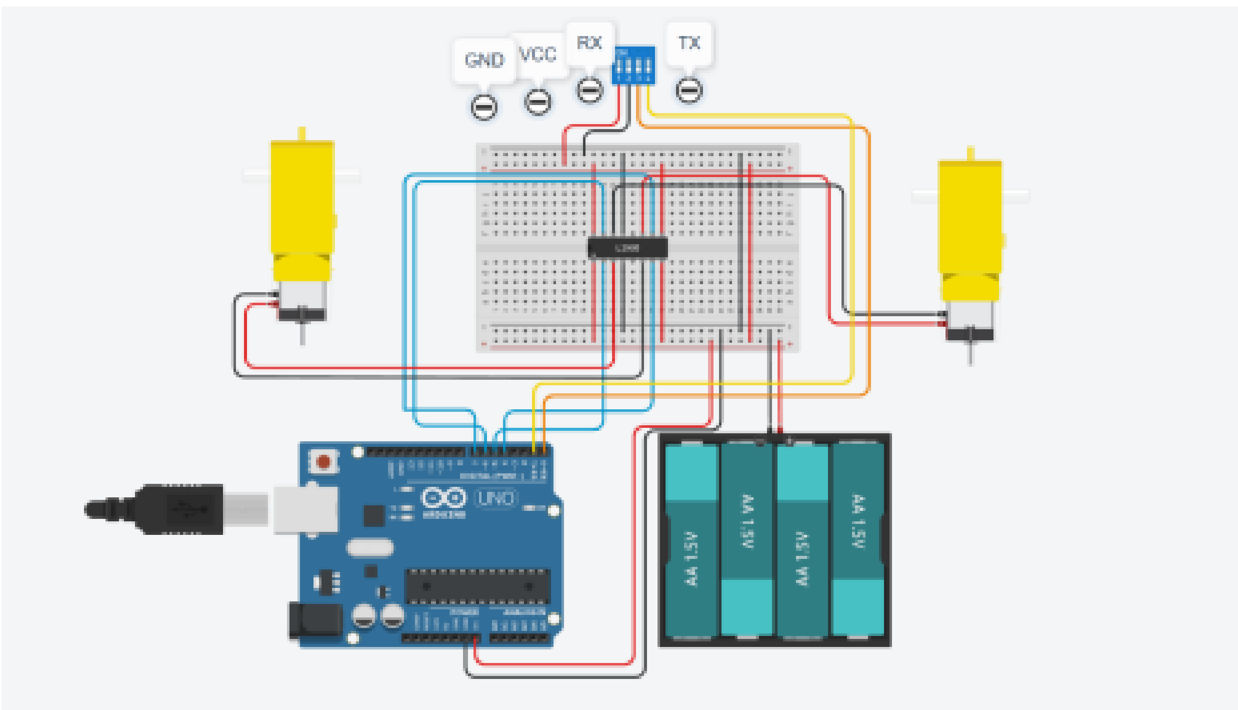

Fig 4.1: Circuit Diagram for Bluetooth Control Robot.

5. Code / Assembly Program

```
char command;

int ENA = 5;
int IN1 = 8;
int IN2 = 9;

int ENB = 6;
int IN3 = 10;
int IN4 = 11;

void setup() {
 Serial.begin(9600);
```

```
 pinMode(ENA, OUTPUT);
 pinMode(IN1, OUTPUT);
 pinMode(IN2, OUTPUT);

 pinMode(ENB, OUTPUT);
 pinMode(IN3, OUTPUT);
 pinMode(IN4, OUTPUT);
 }

void loop() {
```



```
 if (Serial.available()) {
command = Serial.read();  }

 switch(command) {

 case 'F': // Forward
forward();
 break;

 case 'B': // Backward
backward();
 break;

 case 'L': // Left
 left();
 break;

 case 'R': // Right
 right();
 break;

 case 'S': // Stop
 stopRobot();
 break;
 }
 }

void forward() {
 digitalWrite(IN1,HIGH);
digitalWrite(IN2,LOW);
digitalWrite(IN3,HIGH);
digitalWrite(IN4,LOW); }

void backward() {
```

```
 digitalWrite(IN1,LOW);
digitalWrite(IN2,HIGH);
digitalWrite(IN3,LOW);
digitalWrite(IN4,HIGH);
}

void left() {
 digitalWrite(IN1,LOW);
digitalWrite(IN2,HIGH);
```



```
 digitalWrite(IN3,HIGH);
 digitalWrite(IN4,LOW);
 }

void right() {
 digitalWrite(IN1,HIGH);
 digitalWrite(IN2,LOW);
 digitalWrite(IN3,LOW);
 digitalWrite(IN4,HIGH);
 }

void stopRobot() {
 digitalWrite(IN1,LOW);
 digitalWrite(IN2,LOW);
 digitalWrite(IN3,LOW);
 digitalWrite(IN4,LOW);
 }
```

6. Output / Observations

- Bluetooth successfully paired with Android phone.
- Commands sent from mobile app controlled robot movement.
- Robot responded to:

  - **F** → Move Forward
  - **B** → Move Backward
  - **L** → Turn Left
  - **R** → Turn Right
  - **S** → Stop

- Motors operated smoothly using external battery supply.

7. Result

The Bluetooth-based Android-controlled robot was successfully designed and implemented. Wireless communication between the Android device and Arduino UNO was established, enabling real-time robot control.

8. Conclusion

This experiment demonstrated how wireless Bluetooth communication can be integrated with Arduino for robotic control. The L293D motor driver effectively controlled motor direction and speed, while the HC-05 module provided reliable wireless connectivity. The system can be extended for automation, surveillance robots, or smart vehicle applications.